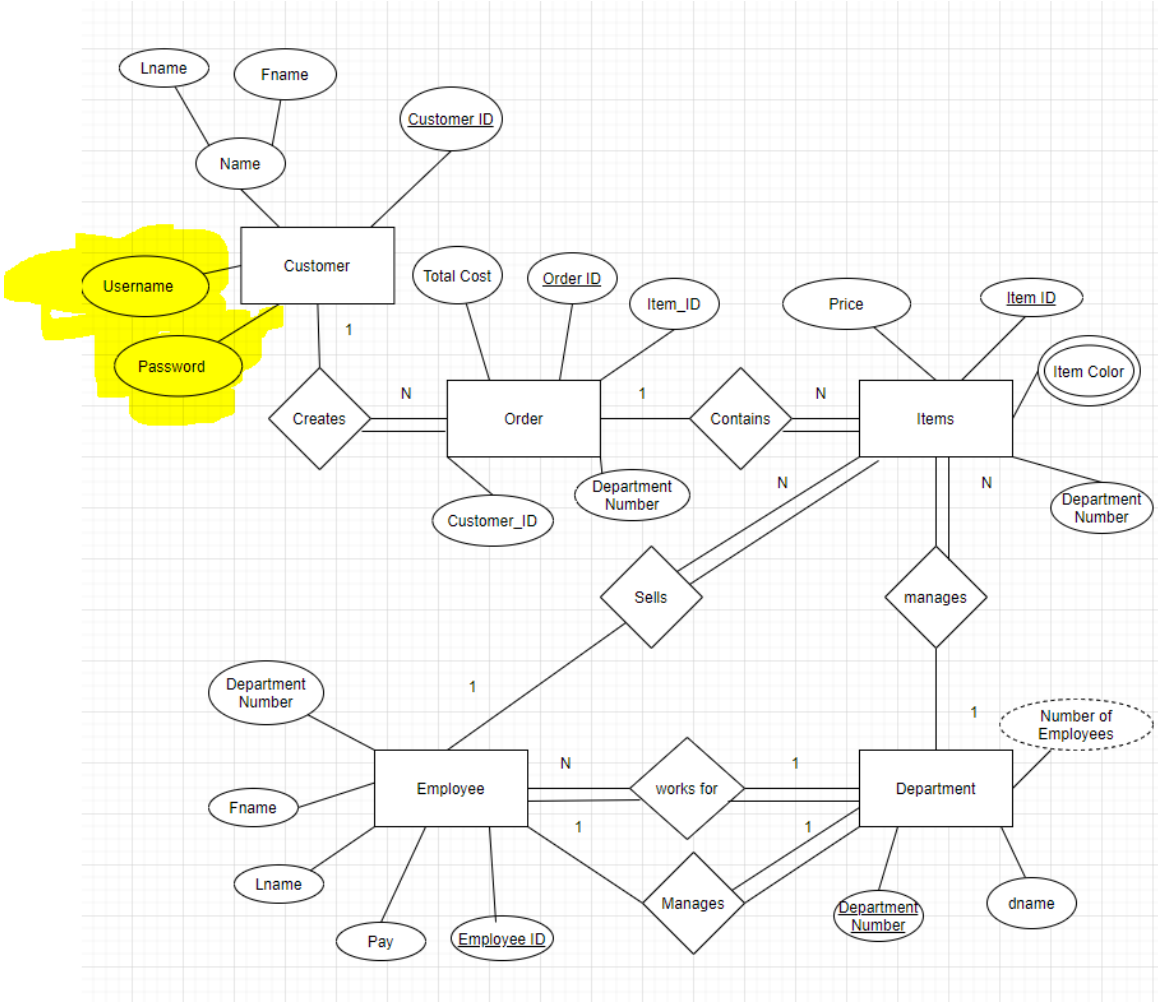
Cloothing Universe!

Jacob Harmon and Andrew Li

Problem Statement and Conceptual Design

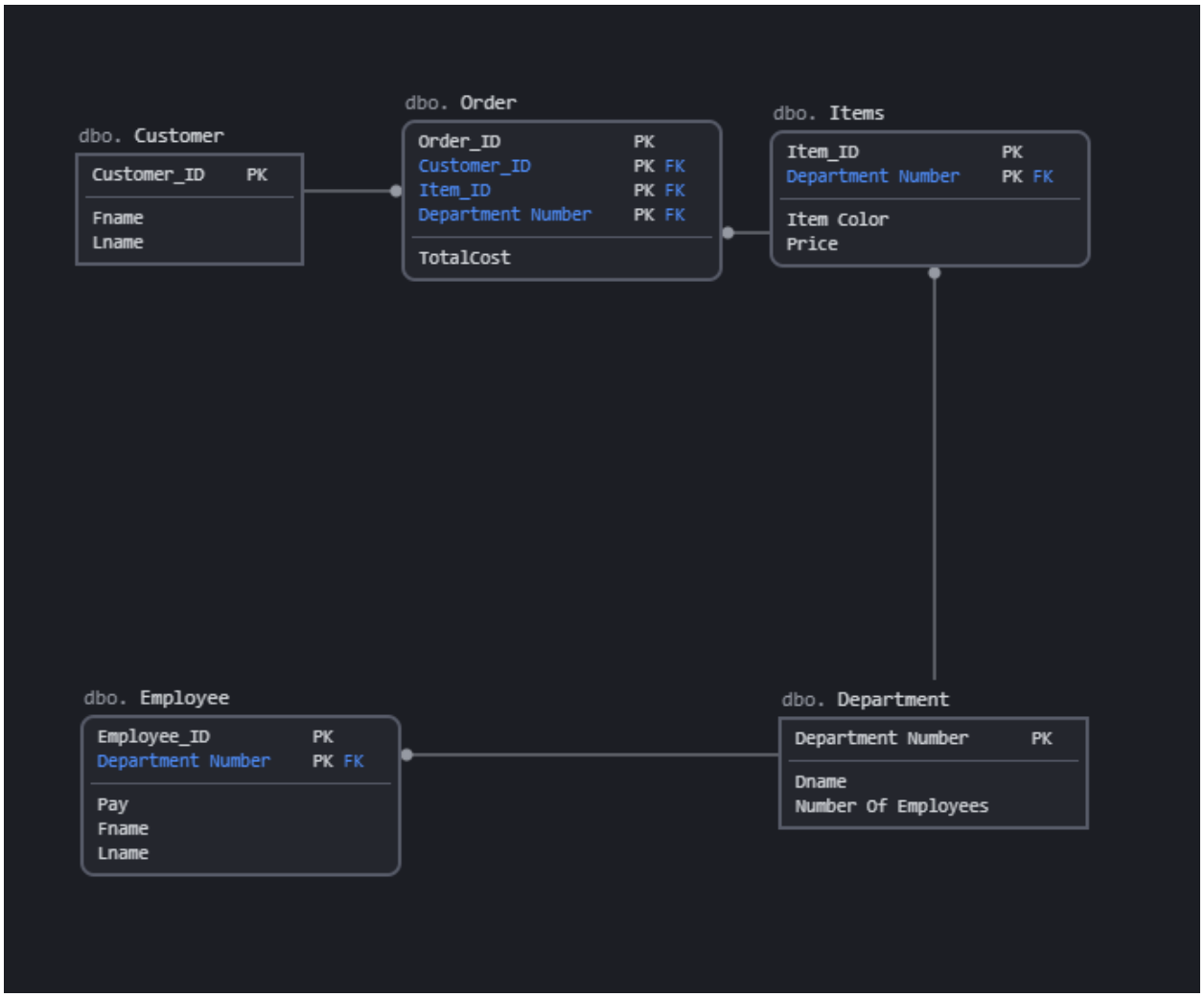
Our project idea is to host an electronic store that sells clothing items. The purpose of this database is to keep track of employees, customers, and clothing items. Regarding the scope of our project, we aim to create a command line program using Python and SQL that provides information about the electronic store. This electronic store will be based off other online stores.



The ER diagram describes a customer creating an online order containing items managed by departments filled with employees who sell them.

The goals of this project is to be able to search and sort the entities. One implementation is to gather a list of clothing items and their information so that these entities can be searchable and sortable by clothing id, clothing type, manufacturer, etc. Another implementation is to be able to search for employees and customers in an efficient manner whether it'll be about which employee works for which department or whether customers are in the rewards program. The entities for this project are employees, departments, items, orders, and customers.

Logical Database Design



(The top part of each table is its primary key, I couldn't figure out how to underline text in this program.)

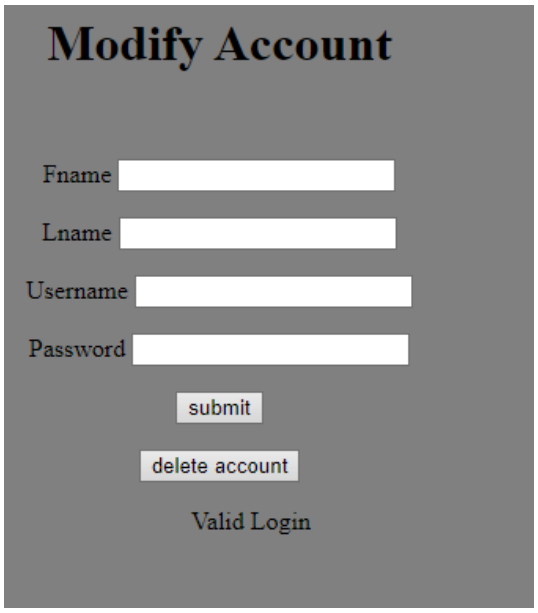
Data Types Table

TABLE	ATTRIBUTE	TYPE	CONSTRAINT
Customer	Customer_ID	INTEGER	Primary Key
Customer	Fname	CHAR(80)	NOT NULL
Customer	Lname	CHAR(80)	NOT NULL
Order	Order_ID	INTEGER	Primary Key
Order	TotalCost	INTEGER	NOT NULL
Order	Customer_ID	INTEGER	Foreign Key
Order	Item_ID	INTEGER	Foreign Key
Order	Department_ID	INTEGER	Foreign Key
Item	Item_ID	INTEGER	Primary Key
Item	Price	INTEGER	NOT NULL
Item	Department Number	INTEGER	Foreign Key
Department	Department Number	INTEGER	Primary Key
Department	Number_of_Empoyees	INTEGER	NOT NULL
Department	Dname	CHAR(80)	NOT NULL
Employee	Employee_ID	INTEGER	Primary Key
Employee	Department Number	INTEGER	Foreign Key
Employee	Pay	INTEGER	NOT NULL
Employee	Fname	CHAR(80)	NOT NULL
Employee	Lname	CHAR(80)	NOT NULL

Application Program Design

```
Login()
        ID
        if(ID = query employee table for match)
                Employee = true
                Logged_in = true
        else if(ID = query customer table for match)
                Customer = true
                Logged_in = true
        Else
                Display failed login message

AddItem()
        if(query item == acceptable)
                Add item to Order
        else
                Display invalid item message

Add_User()
        if(employee = true)
                Specify if employee or customer
                Execute query to add user attributes to customer or employee table

Delete_User()
        if(employee = true)
                Specify user
                Execute delete query at employee or customer table
                Cascade downwards

Modify_User()
        if(employee = true)
                Specify user
                Execute modification query for attributes under customer or employee table

CheckOut()
        Get Order ID from customer table
        Get total cost from order
        Customer pays
        Total cost = 0 for that order
        Delete items from order table
```

UI Sketch

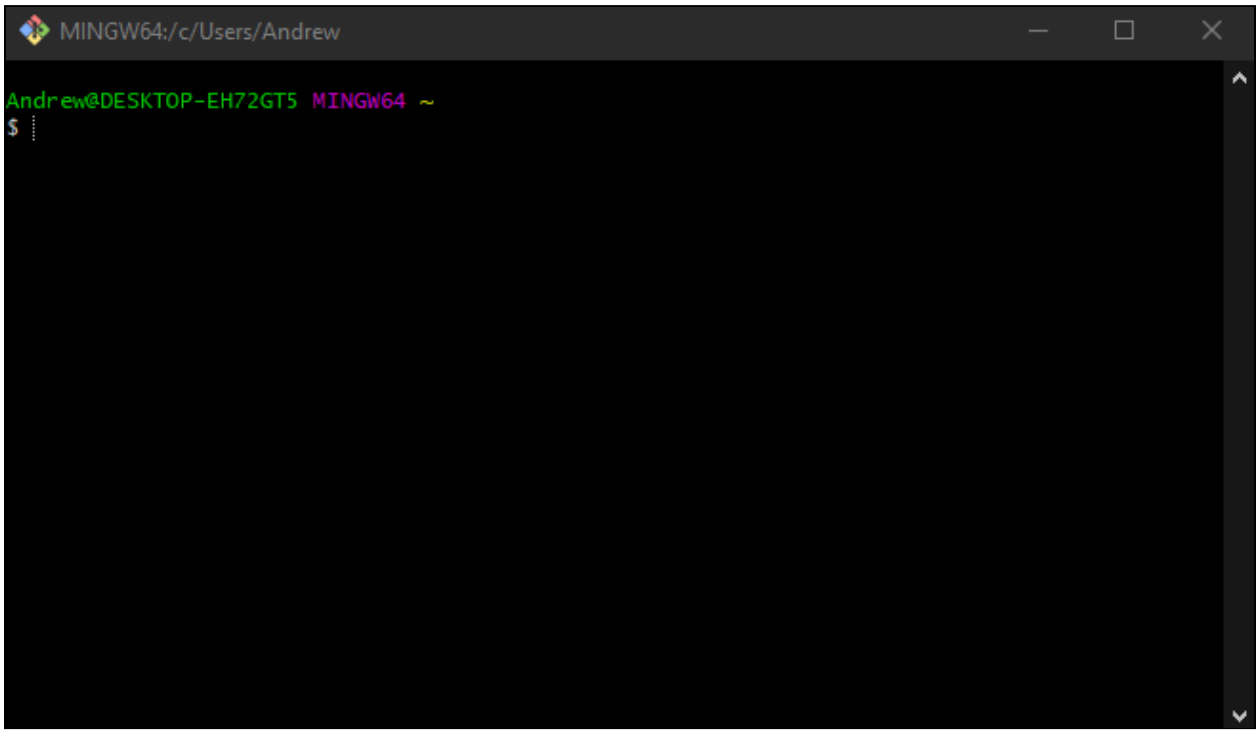


Installation Instructions

This database was created and intended to be run on the Windows operating system (Windows 10). We used MySQL, Python, and Flask to set up the database and frontend. Demonstration and usability will be done on a local host. The following instructions are for making a computer with Python compatible with our application. Before following these instructions, make sure that your computer has Python installed and is configured accordingly.

Installing Python Dependencies

1. Open a Windows command prompt/git bash command prompt.



2. Run the following commands into your command prompt:

   $ pip install Flask

   ```
   Andrew@DESKTOP-EH72GT5 MINGW64 ~
   $ pip install Flask
   ```

   $ pip install mysql

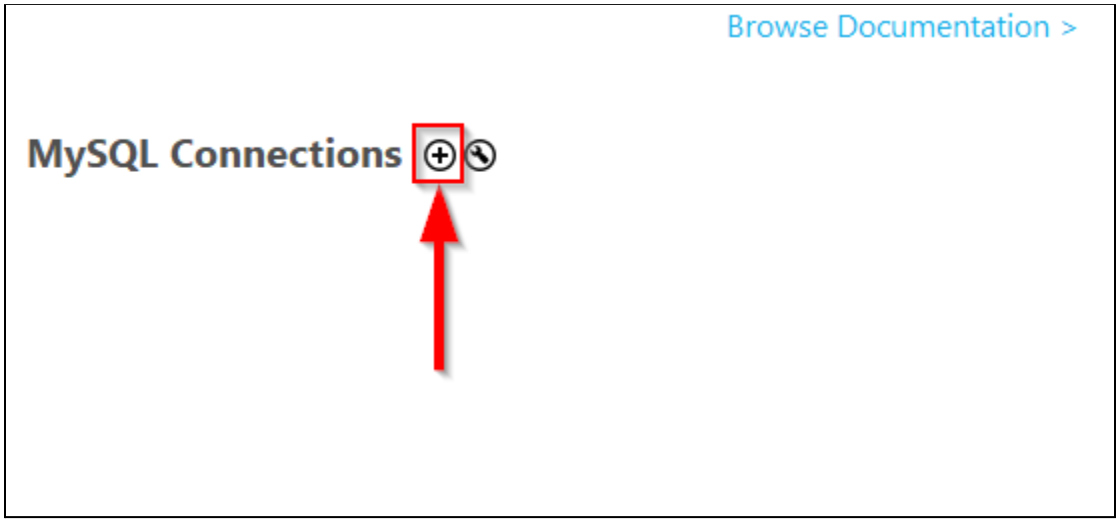
   ```
   Andrew@DESKTOP-EH72GT5 MINGW64 ~
   $ pip install mysql
   ```

   $ pip install mysql-connector-python

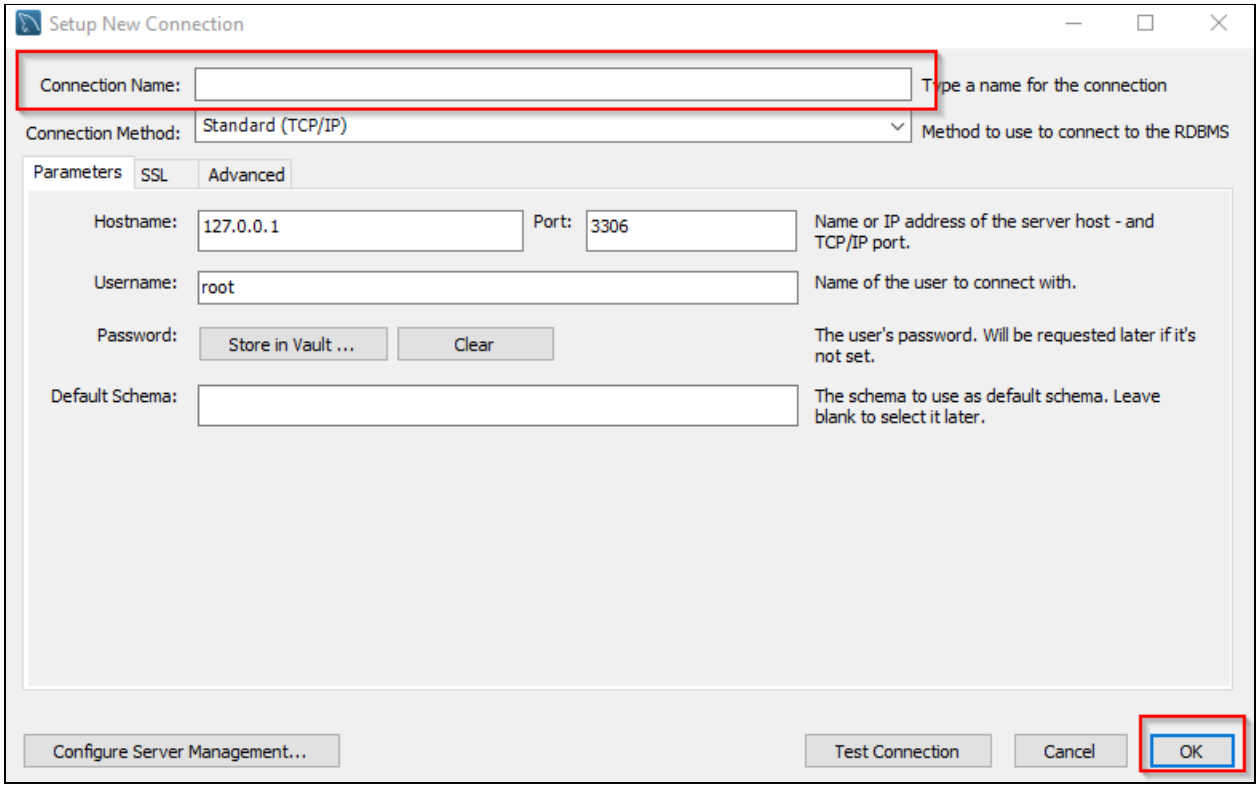
   ```
   Andrew@DESKTOP-EH72GT5 MINGW64 ~
   $ pip install mysql-connector-python
   ```

Creating the database using MySQL workbench

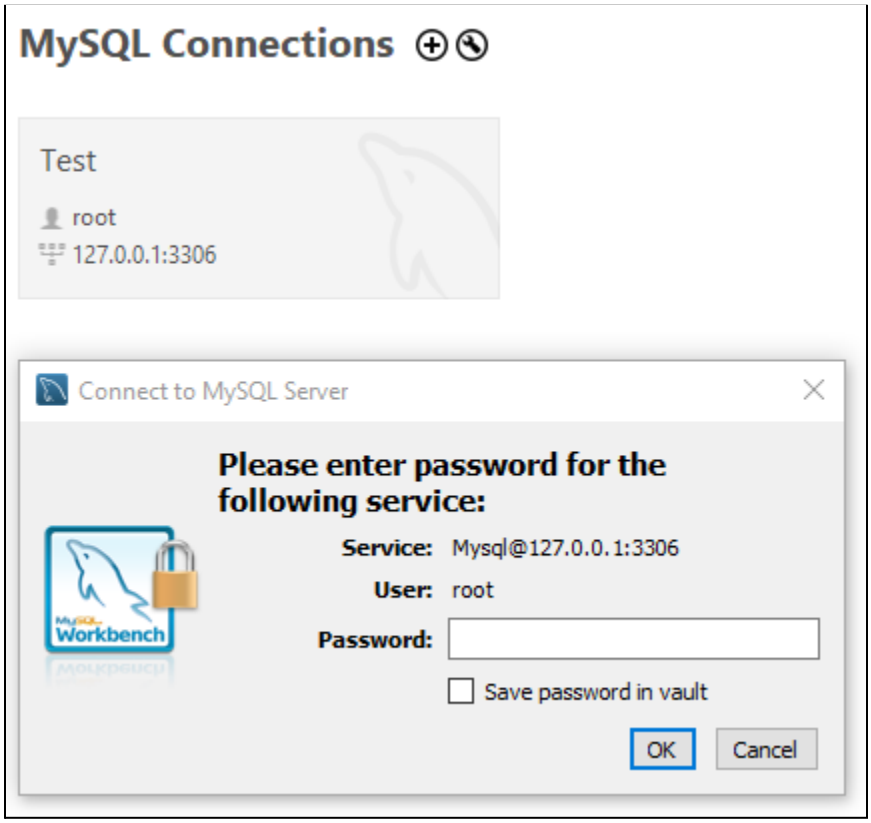
1. Select the small '+' to allow for a new connection.



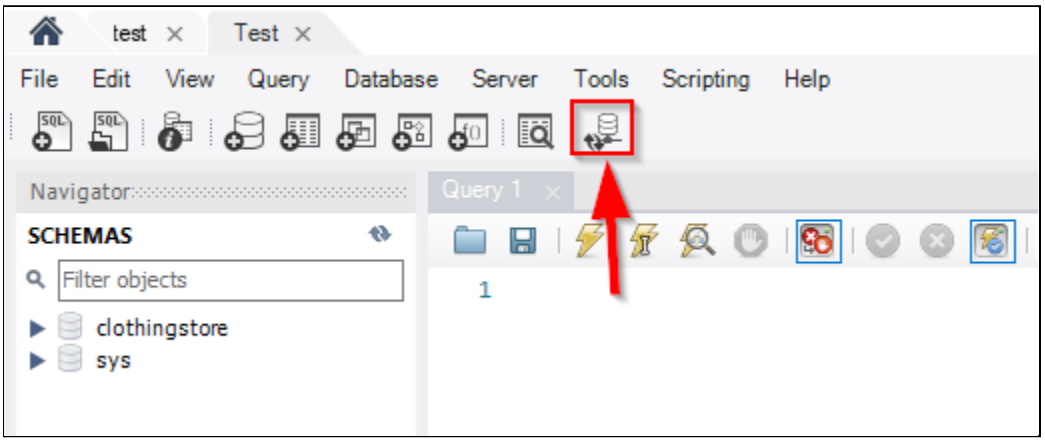
2. Add a name in the "Connection Name" field and then press "OK".



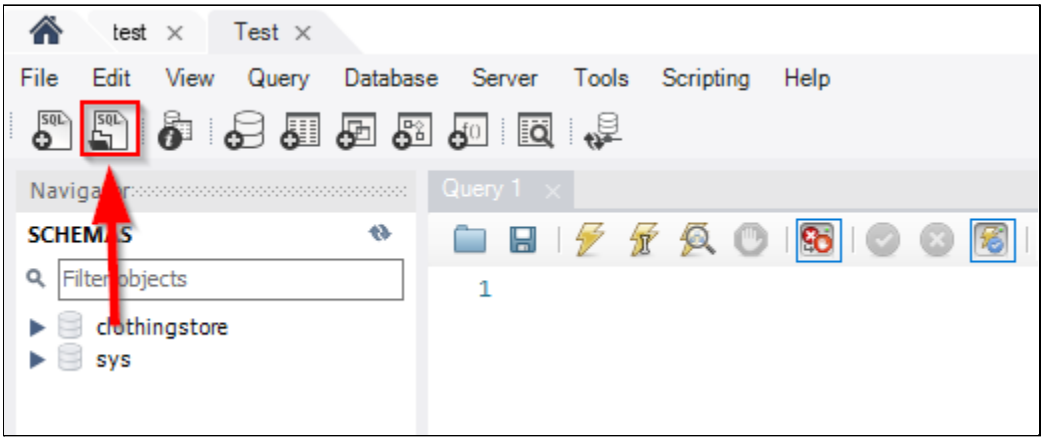
3. Select the newly made connection and enter the correct credentials.



4. Once you are logged in, make sure you are connected to the SQL server. If you are not, make sure to “reconnect the DBMS” at the top of the toolbar.



5. Now that you are connected to the SQL server, you can now “Open a SQL Script file in a new query tab” at the top of the toolbar. Make sure that you select the correct .sql file.



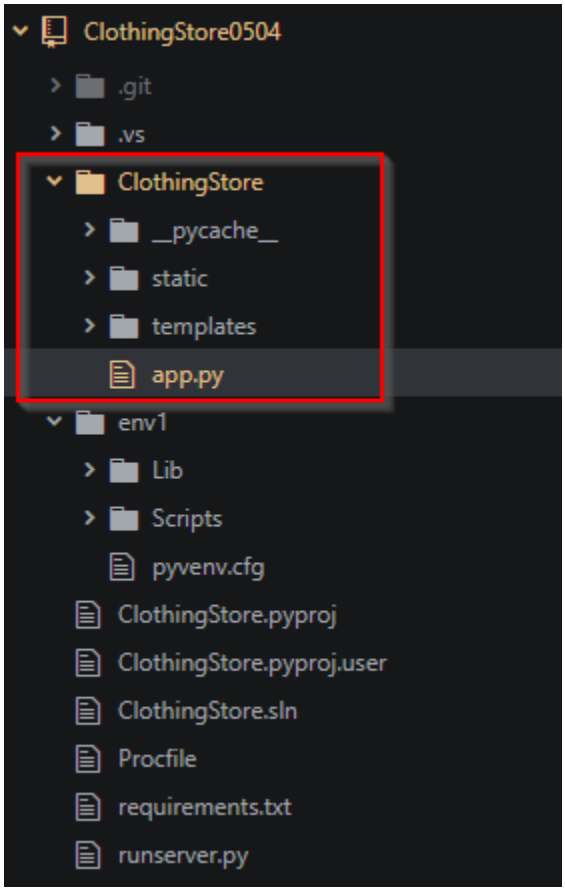
6. Once the .sql file has been opened, you can either select the lightning bolt symbol to run all of the code.

You can also use the lightning bolt with the typing cursor to run the query where your cursor is placed.

Running the Flask App

1. Download the source code provided. You may need to extract the code.

2. Open “app.py” under ClothingStore/app.py.



3. Go to line 20 in “app.py” and change the password to your root user’s password. Save the file once edited.

app.py

```
1  from flask import Flask, render_template, request, flash, redirect, url_for
2  import mysql.connector
3  import random
4  from mysql.connector import Error, connection
5
6  loggeduser = ""
7  loggedpwd = ""
8
9  app = Flask(__name__)
10 app.debug = True
11
12 app.config['SESSION_TYPE'] = 'filesystem'
13 app.config['SECRET_KEY'] = 'super secret key'
14 app.config['MYSQL_HOST'] = 'localhost'
15 app.config['MYSQL_USER'] = 'root'
16 app.config['MYSQL_PASSWORD'] = 'root'
17 app.config['MYSQL_DB'] = 'clothingstore'
18
19 #Database functions
20 cnx = connection.MySQLConnection(user = 'root', password = 'root', host = 'localhost', database = 'clothingstore',
```

4. Open up a git bash or Windows terminal to run the Python app.

MINGW64:/c/Users/Andrew

```
Andrew@DESKTOP-EH72GT5 MINGW64 ~
$
```

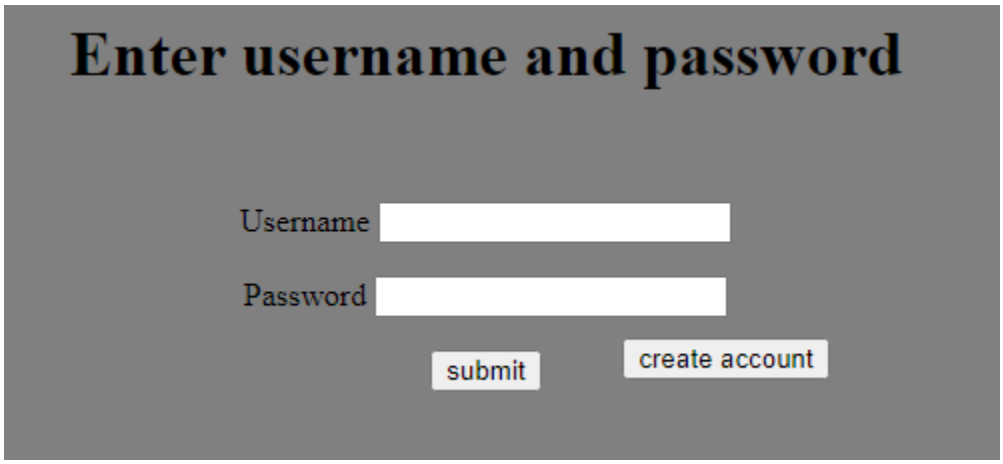
5. Go into the source code directory. Ensure that you are in the “ClothingStore” directory. Adjust accordingly using terminal commands (cd).

```
MINGW64:/c/Users/Andrew/Desktop/ClothingStore0504/ClothingStore

Andrew@DESKTOP-EH72GT5 MINGW64 ~
$ cd Desktop/ClothingStore0504/

Andrew@DESKTOP-EH72GT5 MINGW64 ~/Desktop/ClothingStore0504 (master)
$ cd ClothingStore

Andrew@DESKTOP-EH72GT5 MINGW64 ~/Desktop/ClothingStore0504/ClothingStore (master
)
$
```

6. In the command prompt, run the app.py using the following command: “python app.py” which should display the following output.

```
Andrew@DESKTOP-EH72GT5 MINGW64 ~/Desktop/ClothingStore0504/ClothingStore (master
)
$ python app.py
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 232-897-049
 * Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
127.0.0.1 - - [05/May/2021 21:37:37] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [05/May/2021 21:37:38] "GET /favicon.ico HTTP/1.1" 404 -
```

7. Copy the localhost link and paste it into your browser which should display the homepage of our website.



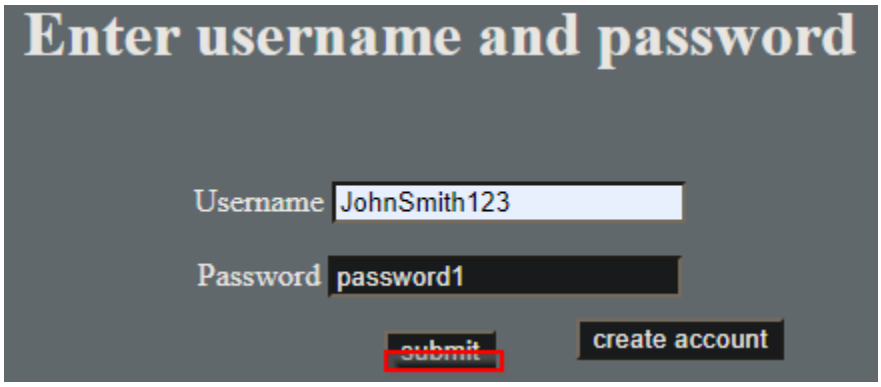
Logging in

1. Make sure that there is an account.

Customer_ID	Fname	Lname	Username	Password
123	John	Smith	JohnSmith123	password1
124	Dinkle	Berg	Dinkle1	123456
125	Crude	Steve	Angryman7	GRRRRRR
126	zygron	aswalla	Alien12	asdfghj
NULL	NULL	NULL	NULL	NULL

Employee 1 | Department 2 | Items 3 | Customer_Order 4 | Customer 5

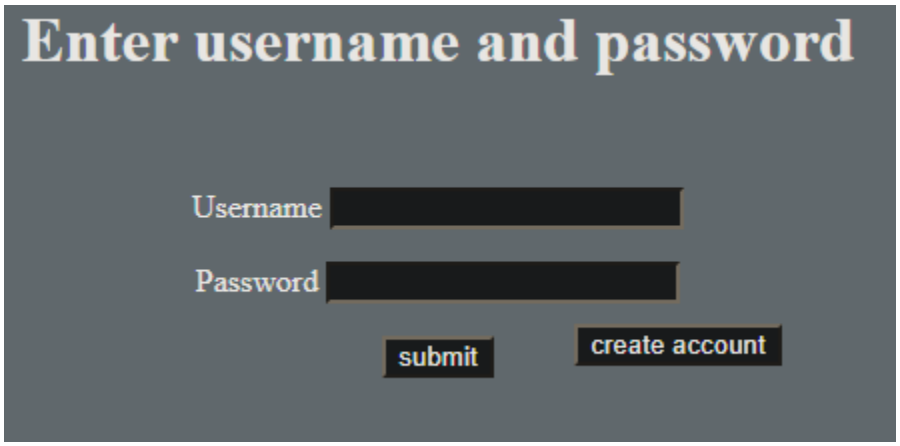
2. Log into the website using a verified account. Click “submit” in the lower button (our html is not lining up).



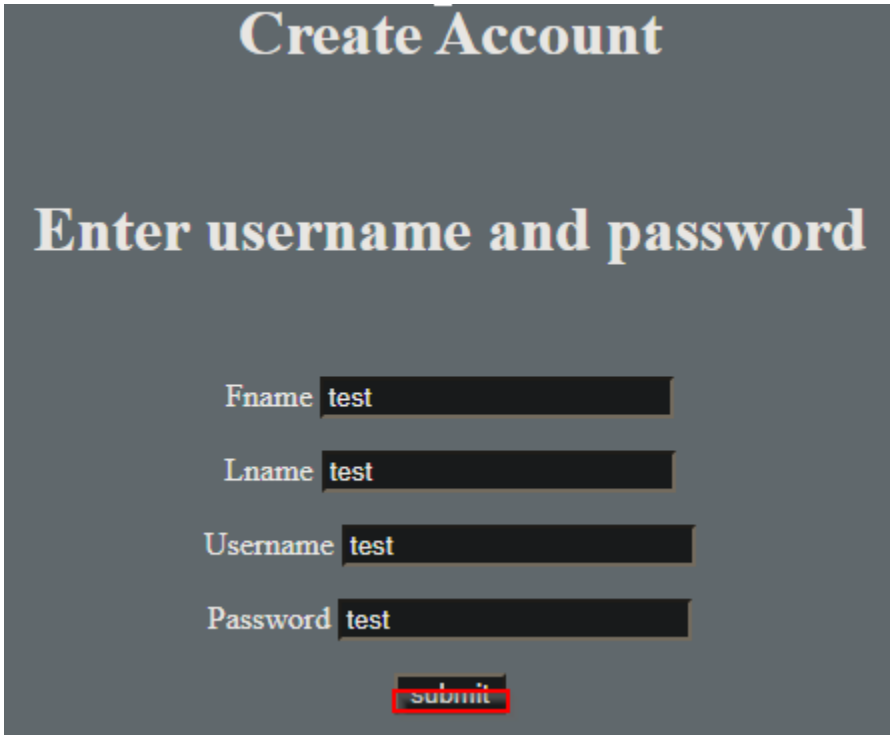
3. It should take you to the “Store Page”.

Creating an account

1. Click on “create account”.



2. Make sure to add a distinct Fname, Lname, Username, and Password. Click “submit” in the lower button (our html is not lining up).

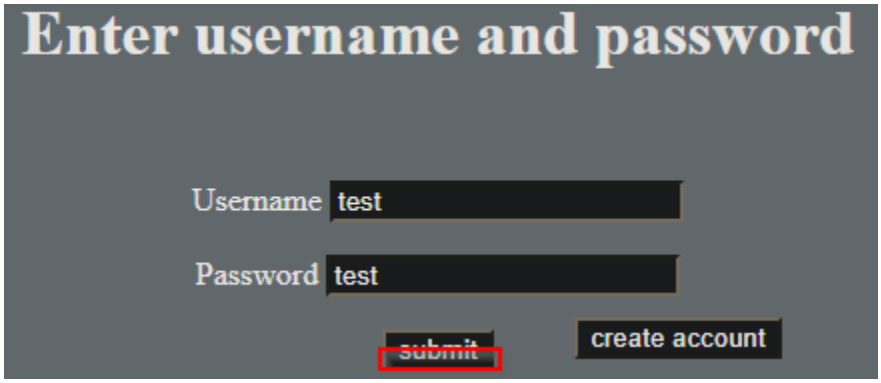


3. Check your database to see if the account is generated.

Customer_ID	Fname	Lname	Username	Password
123	John	Smith	JohnSmith123	password1
124	Dinkle	Berg	Dinkle1	123456
125	Crude	Steve	Angryman7	GRRRRRR
126	zygron	aswalla	Alien12	asdfghj
12111	test	test	test	test
NULL	NULL	NULL	NULL	NULL

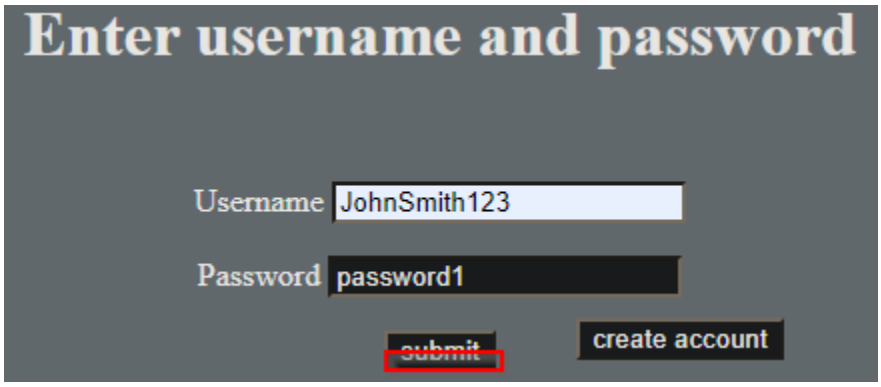
Employee 6 | Department 7 | Items 8 | Customer_Order 9 | Customer 10 ×

4. Log into the page using your new account credentials.



Add items to cart and checkout

1. Log into the page using valid account credentials.



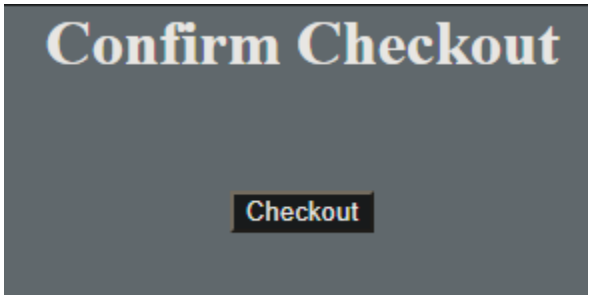
2. Add any item on the “Store Page”.

3. Check to see if your user id matches with the cart items in your account.

Order_ID	Cost	Items_Item_ID	Customer_Customer_ID
1	0	1	123
2	10	2	12111
3	10	2	12111
4	10	2	12111
5	10	2	12111
NULL	NULL	NULL	NULL

4. Click on “checkout”.

5. This should take you to the “Confirm Checkout” page. Click on “Checkout”.

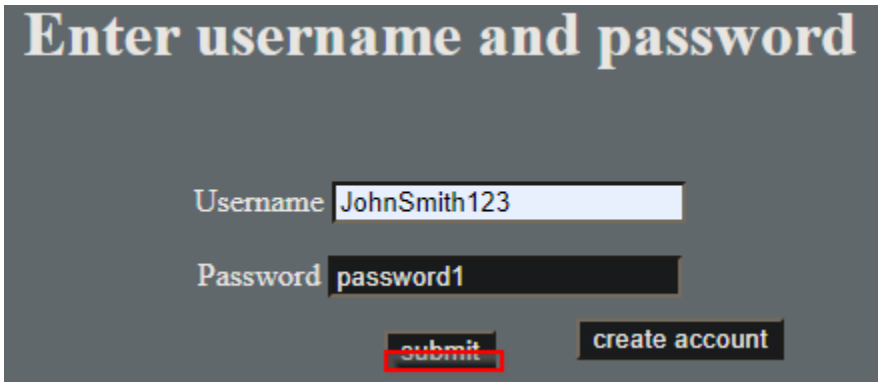


6. After checking out, your account should have no more items in the cart. Open my SQL Workbench to see if there are any items left under your user ID.

Order_ID	Cost	Items_Item_ID	Customer_Customer_ID
1	0	1	123
NULL	NULL	NULL	NULL

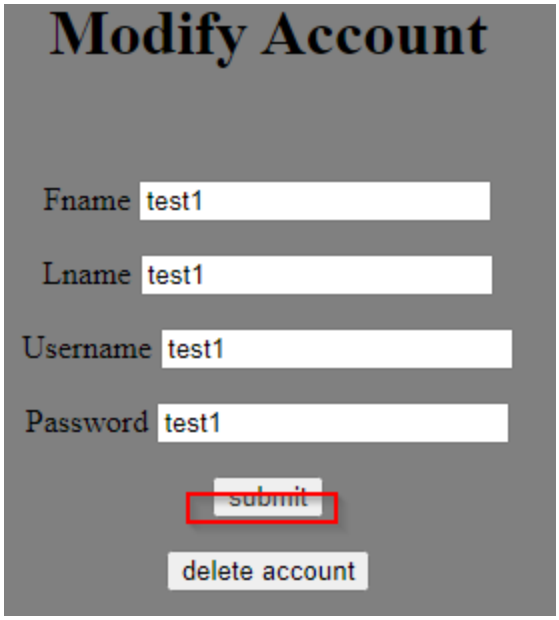
Modify Account

1. Log into the page using valid account credentials.

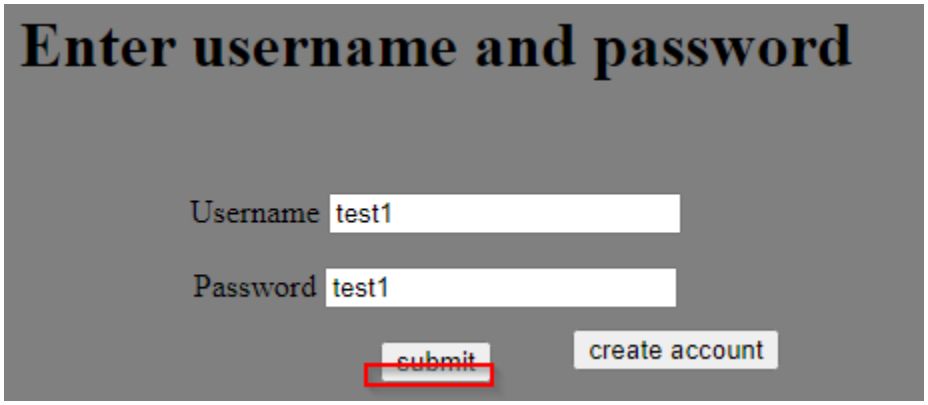


2. Click “modify account”.

3. Make sure to add a distinct Fname, Lname, Username, and Password. Click “submit” in the lower button (our html is not lining up). This will then change the values in the database for your user

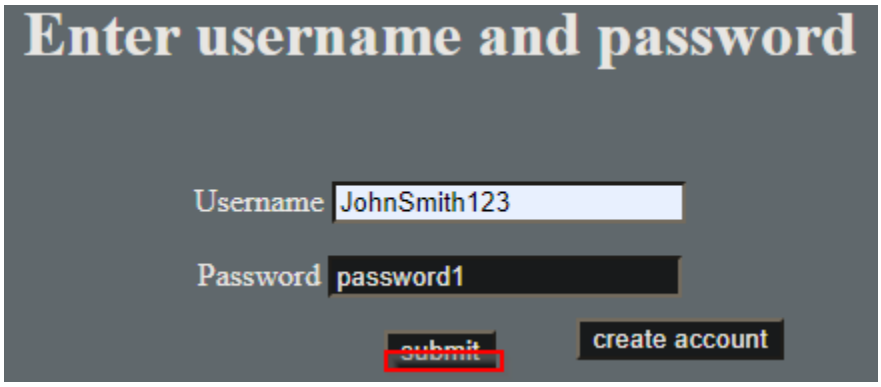


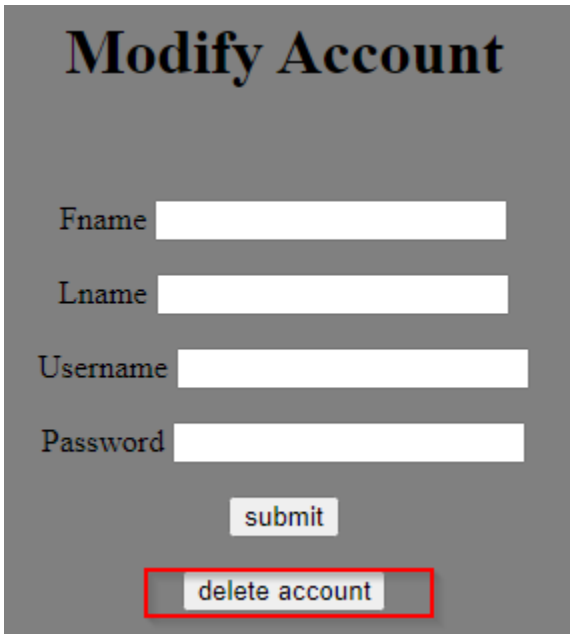
4. Log into the page using your new valid credentials which should take you to the “Store Page”.



Delete account

1. Log into the page using valid account credentials.



2. Click “modify account”.

3. Click on “delete account” to delete the account. NOTE: If the account has items, the items will also be deleted.